

Python Programming

Unit 01 – Lecture 04: Input/Output, Escape Sequences, Operators, Precedence

Tofik Ali

School of Computer Science, UPES Dehradun

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Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

- Read input and convert to correct types

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- Format output using `print()` options and escape sequences

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- Apply operators: arithmetic, relational, logical, bitwise, membership, identity

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- Read input and convert to correct types
- Format output using `print()` options and escape sequences
- Apply operators: arithmetic, relational, logical, bitwise, membership, identity
- Evaluate expressions using precedence and parentheses

Input and Output

- `input(prompt)` reads text from keyboard (returns `str`)

```
a = int(input("Enter a: "))  
b = int(input("Enter b: "))  
print("a =", a, "b =", b, sep=" | ")
```


Input and Output

- `input(prompt)` reads text from keyboard (returns `str`)
- `print(...)` displays output

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b = int(input("Enter b: "))  
print("a =", a, "b =", b, sep=" | ")
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Input and Output

- `input(prompt)` reads text from keyboard (returns `str`)
- `print(...)` displays output
- Useful `print` options: `sep`, `end`

```
a = int(input("Enter a: "))  
b = int(input("Enter b: "))  
print("a =", a, "b =", b, sep=" | ")
```

Escape Sequences

- New line: `\n`

```
print("Line1\nLine2")  
print("A\tB\tC")  
print("C:\\\\Users\\\\\\tofik.ali")
```

Escape Sequences

- New line: `\n`
- Tab: `\t`

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print("Line1\nLine2")  
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Escape Sequences

- New line: `\n`
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- Backslash: `\\`

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print("Line1\nLine2")  
print("A\tB\tC")  
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```

Escape Sequences

- New line: `\n`
- Tab: `\t`
- Backslash: `\\`
- Quotes inside strings: `\"` or `\'`

```
print("Line1\nLine2")  
print("A\tB\tC")  
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```

Operator Categories

Category	Examples
Arithmetic	+ - * / % // **
Relational	< <= > >= == !=
Logical	and or not
Bitwise	& ^ ~ << >>
Membership	in, not in
Identity	is, is not

Precedence (Why Parentheses Matter)

- Multiplication happens before addition

```
print(2 + 3 * 4)      # 14  
print((2 + 3) * 4)    # 20
```


Precedence (Why Parentheses Matter)

- Multiplication happens before addition
- Use parentheses to make intention clear

```
print(2 + 3 * 4)      # 14  
print((2 + 3) * 4)    # 20
```

Membership and Identity

- Membership: check presence in a sequence/collection

```
nums = [10, 20, 30]
print(20 in nums)    # True
```

```
a = [1, 2]
b = a
c = [1, 2]
print(a is b)         # True (same object)
print(a is c)         # False (different objects)
```

Membership and Identity

- Membership: check presence in a sequence/collection
- Identity: check if two names refer to the same object

```
nums = [10, 20, 30]
print(20 in nums)    # True
```

```
a = [1, 2]
b = a
c = [1, 2]
print(a is b)        # True (same object)
print(a is c)        # False (different objects)
```

Demo: Operator Precedence + Truth Tables

- File: `demo/operator_precedence_playground.py`

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- File: `demo/operator_precedence_playground.py`
- Prints examples for:
 - precedence surprises
 - bitwise truth table (`&`, `—`, `^`)
 - membership and identity checks

Checkpoint 1

Question: Evaluate and explain:

- $2 + 3 * 4$
- $(2 + 3) * 4$

Checkpoint 2

Question: What is the difference between `==` and `is`?

Think-Pair-Share

Discuss:

- Why can wrong input types break programs?
- How would you validate age so that it is a non-negative integer?

Key Takeaways

- Convert input strings to numbers using `int/float`

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Key Takeaways

- Convert input strings to numbers using `int/float`
- Escape sequences help print clean output
- Know operator categories and choose the right one
- Precedence matters; parentheses improve correctness and readability

Exit Question

Give one example each:

- membership operator
- identity operator